AI-Assisted Evidence Linking: A Strategy to Improve Research Use in Polarized K-12 Education Policymaking

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Research Questions

We address three interconnected research questions at the intersection of AI technology and evidence-based education policymaking:

RQ1: How do policymakers reference scientific evidence in their public communications about polarized K-12 education issues, and does the accuracy of these references differ systematically by party affiliation?

RQ2: Can AI systems reliably identify empirical claims in policy communications, retrieve relevant scientific literature, and evaluate the degree of scientific support for those claims?

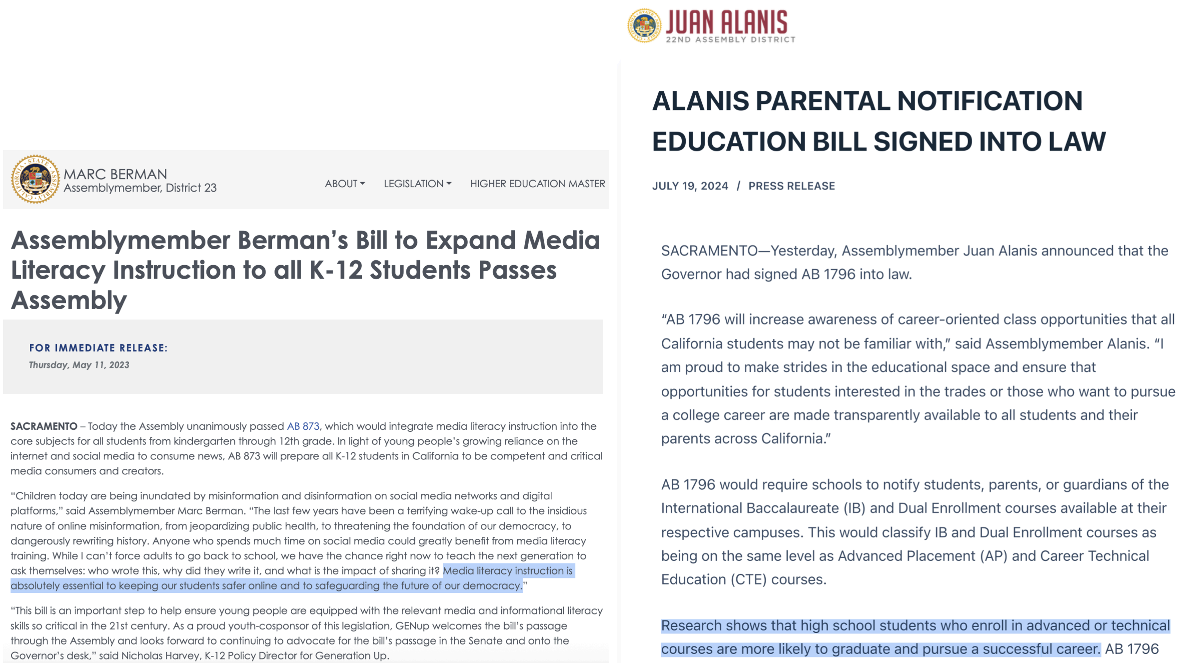
RQ3: Does providing legislators with AI-synthesized evidence reports improve their use of research evidence in education policymaking?

Rationale and Objective

Education policy has become one of the most politically polarized domains in American governance (DeBray 2006, Houston 2024, Collins and Reckhow 2024). Debates over social-emotional learning (SEL), school safety measures, reading instruction methods, and educational technology increasingly divide along partisan lines, with competing claims about evidence (Joslyn and Haider-Markel 2014). This polarization creates both challenges and opportunities for improving research use: Policymakers may cite scientific evidence strategically to support predetermined positions, yet they also face genuine information asymmetries when navigating complex literatures spanning child development, educational psychology, and implementation science (Brown 2014, Lubienski et al. 2014).

Recent advances in large language models (LLMs) and retrieval-augmented generation (RAG) create unprecedented opportunities to bridge the gap between scientific knowledge and policy discourse. These technologies can identify empirical claims in natural language, search vast scientific databases, and synthesize evidence in accessible formats (Koneru et al. 2024, Wang and Clark 2024). However, their application to policymaking communication remains unexplored. In this project we develop and test an artificial intelligence-assisted evidence linking system specifically designed to improve research use in polarized contexts..

The core technical objective of this project is to develop an AI-assisted evidence linking system that processes policymaking documents through a structured pipeline. The system (1) ingests policy communications such as press releases, bill analyses, and committee testimony; (2) identifies claims within these documents that invoke or should be supported by scientific evidence; (3) retrieves relevant peer-reviewed literature and evaluates the degree to which the scientific record supports, refutes, or is silent on each claim; and (4) surfaces these assessments in accessible “Evidence Reports” designed for time-constrained policymakers. By automating the labor-intensive process of linking policy claims to scientific evidence, this system addresses a key barrier to research use identified in prior work: policymakers' limited time and capacity to locate and evaluate relevant research (Langer et al. 2016, Bogenschneider et al. 2019).



MARC BERMAN
Assemblymember, District 23

ABOUT LEGISLATION HIGHER EDUCATION MASTER

Assemblymember Berman's Bill to Expand Media Literacy Instruction to all K-12 Students Passes Assembly

FOR IMMEDIATE RELEASE:
Thursday, May 11, 2023

SACRAMENTO – Today the Assembly unanimously passed AB 873, which would integrate media literacy instruction into the core subjects for all students from kindergarten through 12th grade. In light of young people's growing reliance on the internet and social media to consume news, AB 873 will prepare all K-12 students in California to be competent and critical media consumers and creators.

"Children today are being inundated by misinformation and disinformation on social media networks and digital platforms," said Assemblymember Marc Berman. "The last few years have been a terrifying wake-up call to the insidious nature of online misinformation, from jeopardizing public health, to threatening the foundation of our democracy, to dangerously rewriting history. Anyone who spends much time on social media could greatly benefit from media literacy training. While I can't force adults to go back to school, we have the chance right now to teach the next generation to ask themselves: who wrote this, why did they write it, and what is the impact of sharing it? Media literacy instruction is absolutely essential to keeping our students safer online and to safeguarding the future of our democracy."

"This bill is an important step to help ensure young people are equipped with the relevant media and informational literacy skills so critical in the 21st century. As a proud youth-cosponsor of this legislation, GENup welcomes the bill's passage through the Assembly and looks forward to continuing to advocate for the bill's passage in the Senate and onto the Governor's desk," said Nicholas Harvey, K-12 Policy Director for Generation Up.

JUAN ALANIS
22ND ASSEMBLY DISTRICT

ALANIS PARENTAL NOTIFICATION EDUCATION BILL SIGNED INTO LAW

JULY 19, 2024 / PRESS RELEASE

SACRAMENTO—Yesterday, Assemblymember Juan Alanis announced that the Governor had signed AB 1796 into law.

"AB 1796 will increase awareness of career-oriented class opportunities that all California students may not be familiar with," said Assemblymember Alanis. "I am proud to make strides in the educational space and ensure that opportunities for students interested in the trades or those who want to pursue a college career are made transparently available to all students and their parents across California."

AB 1796 would require schools to notify students, parents, or guardians of the International Baccalaureate (IB) and Dual Enrollment courses available at their respective campuses. This would classify IB and Dual Enrollment courses as being on the same level as Advanced Placement (AP) and Career Technical Education (CTE) courses.

Research shows that high school students who enroll in advanced or technical courses are more likely to graduate and pursue a successful career. AB 1796

Figure 1. *Education Policy Press Releases from Democrat (Berman) and Republican (Alanis) CA Reps. Technical claims highlighted.*

Policy Communication in Press Releases: The Case of CA State Legislators

Lawmakers communicate with constituents, media, and stakeholders through official press releases issued by their offices. These documents serve multiple functions in the policymaking process: they announce legislative priorities and bill introductions, frame policy problems and proposed solutions, claim credit for policymaking accomplishments, and signal positions on salient issues (Grimmer 2013, Lipinski 2009). Press releases represent a strategic communication channel through which legislators shape public discourse and build support for their policy agendas. Importantly for our purposes, legislators frequently invoke technical evidence in these communications to bolster the credibility of their positions, claiming, for instance, that a proposed intervention "is backed by research" or citing statistics to establish the severity of a problem. Press releases provide direct insight into how policymakers characterize the evidentiary basis for their policy stances. This makes them an ideal corpus for examining patterns in how policymakers reference scientific evidence and for developing AI tools to assess the accuracy of such claims.

We have assembled a substantial corpus of state legislative communications to establish feasibility and inform our research design. Our preliminary dataset comprises 5,131 press releases from California State Assembly members, including 4,279 from Democratic legislators and 852 from Republican legislators. The partisan imbalance is attributable to the fact that approximately 75% of CA legislators are Democrats. This corpus provides a robust foundation for developing and validating our AI pipeline before expanding to a larger sample of states and other document types.

Keyword-based filtering identified 611 education-related press releases (11.9% of the corpus). Notably, Democratic legislators devoted a significantly higher proportion of their communications to education topics (13.1%) compared to Republican legislators (5.9%), suggesting differential policy attention that our full analysis will examine in greater depth. These education-focused communications address a variety of specific issues, including school funding, teacher preparation, student mental health services, and academic standards.

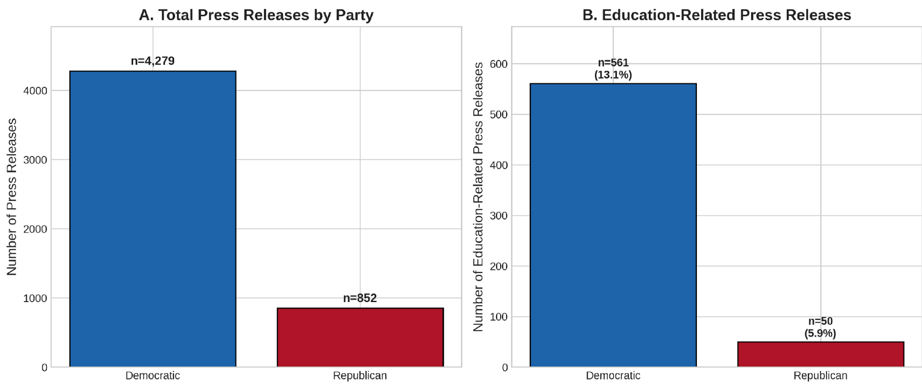


Figure 1. *Corpus composition. Panel A shows total press releases by party. Panel B shows education-related press releases, with Democrats producing 561 (13.1%) and Republicans 50 (5.9%).*

Analysis of polarized topic coverage reveals partisan differences in policy attention. Democratic legislators produced 80 press releases addressing social-emotional learning and student mental health, compared to just 1 from Republican legislators. Similarly, school safety communications were more prevalent among Democrats (n=38) than Republicans (n=4). Reading and curriculum issues showed more balanced attention (Democrats: 26, Republicans: 7), though when normalized by total output (Figure 3), Republicans devoted a slightly higher proportion of communications to this topic (0.82% vs. 0.61%). These patterns suggest that our AI pipeline will encounter substantively different scientific claims depending on the political context—a key consideration for system validation.

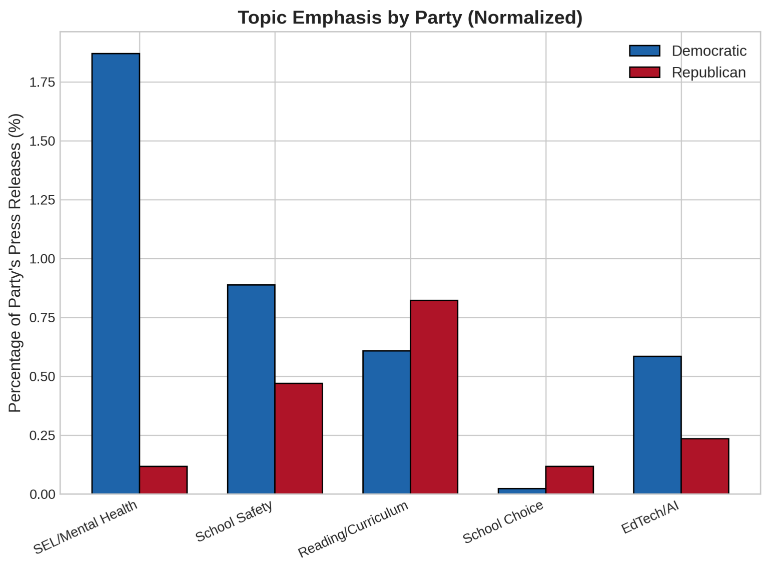


Figure 3. *Topic emphasis normalized by party output. When controlling for overall communication volume, Democrats emphasize SEL/mental health 15 times more than Republicans (1.87% vs. 0.12%), while Republicans show relatively greater attention to reading/curriculum (0.82% vs. 0.61%).*

Theory of Change

Our theory of change posits three mechanisms through which AI-assisted evidence linking can improve research use in polarized contexts. First, by providing legislators with comprehensive, neutrally-presented evidence reports, the intervention may facilitate *instrumental use*—the direct application of research findings to inform specific policy decisions. Second, by exposing legislators to bodies of evidence they might not otherwise encounter, the intervention may promote *conceptual use*—gradual shifts in how legislators understand education issues and evaluate policy options. Third, by providing a common evidentiary foundation, AI-generated reports may support *relational use*—enabling productive dialogue across political divides through shared reference points.

This framework draws on Weiss's (1979) foundational typology while incorporating contemporary insights about embedded research use in policy systems (Oliver et al., 2022). The polarized context creates both barriers and opportunities: legislators may resist evidence that challenges partisan positions, yet they also face constituent and media pressure to justify policy stances. Our design tests whether AI-assisted evidence delivery can navigate these dynamics more effectively than traditional research dissemination approaches.

Methods

Phase 1: Baseline Assessment (Months 1-12)

Building on our preliminary corpus, we will compile a comprehensive dataset of state legislative communications on education policy from as many states as possible, targeting approximately 25,000 press releases over the 2020-2025 period. For most states, it is possible to use conventional web-scraping pipelines to collect the press releases. Our AI pipeline will process these communications through three stages: (1) LLM-based claim extraction using domain-agnostic prompting to identify verifiable scientific claims; (2) parallel search across Crossref, Semantic Scholar, and targeted web sources to retrieve potentially relevant literature; and (3) scientific hypothesis evidencing (Koneru et al. 2024) to determine whether retrieved evidence supports, refutes, or is uninformative regarding each claim. Human coders will validate a stratified sample of at least 1,000 claim-evidence pairs across parties and topics to establish reliability benchmarks.

Phase 2: Tool Development and Pilot Testing (Months 10-24)

We will refine the AI pipeline to generate accessible "Evidence Reports" summarizing the scientific consensus and areas of uncertainty for specific policy claims. These reports will present evidence in a balanced, non-partisan format designed for policymaking professionals. We will pilot the system with at least 50 legislative staff members recruited through email outreach, drawn from 10 states representing diverse political contexts. Participants will receive Evidence Reports on at least three education policy topics being considered at the time of the pilot testing, and complete pre-post surveys measuring research use attitudes and behaviors. Semi-structured interviews with 20 participants will illuminate the mechanisms through which the tool affects (or fails to affect) evidence consideration.

Phase 3: Randomized Evaluation (Months 18-36)

The primary evaluation will employ a randomized controlled trial with at least 2,000 state legislators and senior staff members from several states. Treatment group participants will receive AI-generated Evidence Reports on contentious education policy topics over a six-month period, with reports delivered through email. Control group participants will not receive any email reports. Randomization will be stratified by party, chamber, and state. Any state legislator in a state where education policy is being considered and discussed extensively enough for us to formulate an evidence report will be a subject in the experiment.

Data Analysis

Primary outcomes include: (1) accuracy of scientific citations in subsequent press releases, measured through our AI pipeline and validated by human coding; (2) self-reported research use behaviors on validated instruments; and (3) attitudes toward scientific evidence in education policy. We will employ difference-in-differences estimation to assess treatment effects, with heterogeneity analyses examining whether effects differ by party affiliation, baseline research use, or state political context.

Expected Contributions

This project will generate three primary contributions. First, the baseline assessment will produce the most comprehensive analysis to date of how state legislators communicate about scientific evidence on education policy, with systematic comparisons across parties, states, and policy domains. Second, we will develop and validate an AI system capable of linking policy claims to scientific evidence—a tool with applications well beyond this project. Third, the randomized evaluation will provide causal evidence on whether and how AI-assisted evidence delivery can improve research use in polarized contexts.

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